

Darwix AI — AI Engineer Assessment

Production-Ready Knowledge Base, Grounded Voice Agents & Real-Time Call Intelligence

CANDIDATE

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GITHUB REPOSITORY

CASPER0022/Darwix-Assignment

TEST SUITE

235/235 PASSING (100% Offline)

1. Executive Summary & Verifiable Results

This submission implements a **unified voice AI and real-time intelligence platform** solving all four deliverables across Indian lending, Southeast Asian financial markets, and live contact-center assistance. Every benchmark number below is produced by automated test harnesses from committed real audio and transcript data.

Core Architecture Principle: "The model decides what the customer meant; deterministic code decides everything else." Legal disclosures, loan qualification thresholds, numeric grounding, and escalation policies are strictly owned by code with unit tests.

EVALUATION METRIC	MEASURED RESULT	BENCHMARK SOURCE & EVIDENCE
Q2 Knowledge Base Retrieval	14 Correct, 1 Partial, 0 Incorrect (15 queries)	Pre-declared test set in <code>config/retrieval_tests.yaml</code>
Q4 Live Nudge End-to-End Latency	854 ms p50 / 1,228 ms p95	1.0x wall-clock replay in <code>evaluation/latency_report.md</code>
Q3 Multilingual ASR WER	8.8% (en-IN) · 9.5% (Taglish) · 11.9% (id-ID) · 25.0% (Javanese)	Dialect benchmark in <code>evaluation/asr_benchmark.md</code>
Indexed Knowledge Records	352 searchable (39 PII-redacted & quarantined)	SQLite database + 768d NumPy embeddings
Q1 Turn Latency (to decision)	967 ms p50 (ASR 335 ms + Understand/Slots 632 ms)	Per-turn audit in <code>data/transcripts/sim_q1_*.json</code>
Automated Unit Test Suite	235 Passing (Zero API keys required)	Offline pytest suite in <code>tests/</code>

2. Deliverables & Question-by-Question Overview

Q1: Grounded Voice Agent

en-IN NBFC

Use Case: Small business loan qualification for an Indian NBFC.

Key Features: Browser webcall UI, adaptive VAD with 300ms pre-roll, 5-gate anti-hallucination grounding, strict numeric check, code-enforced legal disclosures, CSV rules data engine, CRM lead persistence.

Q2: Production Knowledge Base

RRF Hybrid

Use Case: Ingestion of real listed NBFC (UGRO Capital) site + policy PDFs.

Key Features: Frequency boilerplate stripping, atomic rule chunking, PII redaction (PAN/Verhoeff Aadhaar), hybrid BM25 + Gemini Dense search fused via Reciprocal Rank Fusion (\$k=60\$).

Q3: Multilingual Voice Bots

PH & ID Locales

Use Case: Philippines life insurance (Taglish) & Indonesia multifinance (Bahasa).

Key Features: Decoupled locale packs, Tagalog honorifics (po /opo), payment vocabulary (GCash/Indomaret), deterministic indirect refusal detection ("Nanti saya kabari deh"), ASR WER benchmark.

Q4: Live Call Nudges

< 1s In-Call

Use Case: Real-time agent assist during live contact center calls.

Key Features: 1.0x wall-clock replay, stereo channel split (Agent L / Customer R), two-tier signals (sub-ms rules + debounced LLMs), nudge cooldowns and duplicate suppression, live WebSocket dashboard.

3. Key Architecture & Grounding Decisions

The 5 Strict Grounding Gates (Anti-Hallucination Pipeline)

When a customer asks a factual question, the response must pass 5 consecutive checks before the agent speaks:

1. **Retrieve First:** No factual claim without an indexed retrieval match.
2. **Confidence Floor:** If top RRF score < 0.35, the agent immediately admits ignorance and offers a human.
3. **Explicit Model Decline:** The LLM is explicitly allowed and instructed to return `false`.
4. **Citation Verification:** All returned `record_id`s must exist in the retrieved chunks.
5. **Strict Numeric Guard:** *Every number in the generated response must appear verbatim in the source chunk*, explicitly preventing loan rate smearing (e.g. turning "14.5% to 16.0%" into "around 15%").

Why No Vector Database?

With 352 searchable records, brute-force cosine similarity over a 768-dimensional NumPy float32 matrix executes in **< 0.2 ms**—three orders of magnitude faster than an ASR API call. Introducing external vector databases (Chroma/Pinecone) would add unnecessary operational overhead with zero latency benefit.

4. Critical Engineering Hurdles Solved

CHALLENGE DISCOVERED	ROOT CAUSE TRACED	ROBUST ENGINEERING SOLUTION
Whisper Hallucinating "Thank You." on Silence	In quiet rooms, the adaptive noise floor decayed from 0.0229 to 0.0012. Soft breathing triggered VAD; Whisper transcribes near-silence into subtitle training artifacts.	1) Clamped noise floor bounds. 2) Added absolute audio energy gate. 3) Subtitle artifact filter across 10 languages.
Noise Floor Latching in Loud Rooms	In loud environments (> 0.013 RMS), continuous tone was classified as speech, freezing the adapter and locking VAD permanently open.	Rewrote noise estimation to sample a low percentile over a rolling 15-second sliding window.
Acoustic Feedback / Self-Talk	Microphone picked up synthesized agent audio from speakers, creating a self-loop.	Elevated VAD threshold dynamically during agent audio playback. Quiet bleed is ignored; loud user barge-in is preserved.
Indonesian Indirect Refusals	Polite phrasing (" <i>Nanti saya kabari deh</i> " — "I'll let you know later") was heard by LLMs as a promise to pay, corrupting collections queues.	Engine detects cultural indirect refusal patterns deterministically in code.

5. Quick Start & Local Verification

The entire system is self-contained with no external dependencies required for testing:

```
# 1. Setup & install (Python 3.11+)
git clone https://github.com/CASPER0022/Darwix-Assignment.git
cd Darwix-Assignment && pip install -e ".[scrape,dev]"

# 2. Run the 235 unit tests (Zero API keys required, 100% offline)
pytest

# 3. Launch interactive web applications
python -m darwix.server.app
```

```
# Open http://localhost:8000 in your browser:  
# → /webcall : Interactive voice agent with live speech & citation inspector  
# → /kb : Knowledge base search with BM25/Dense scoring breakdown  
# → /dashboard : Live agent assist dashboard with real-time audio playback
```